WILP, BITS Pilani

Session 11A: Two-Way ANOVA, Correlation & Regression

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Introduction to Two-Way ANOVA

Isolate two independent factor effects at once without interference.

When analyzing experimental data, we often test how two categorical factors affect a continuous response. Two-Way Analysis of Variance (ANOVA) separates total variation into row factor effects, column factor effects, and unexplained random error.

Everyday Picture & Intuition

Imagine testing baked cake height. One factor is oven temperature (low, medium, high). The second factor is sugar brand (Brand A, Brand B). Two-Way ANOVA tells us if oven temperature matters, if sugar brand matters, and keeps them from confusing each other.

1.1 Mathematical Framework

Total variation (SST) is partitioned into sum of squares for row factor (SSA), column factor (SSB), and error (SSE):

$$SST = SSA + SSB + SSE \tag{1}$$

The degrees of freedom partition accordingly: $rn - 1 = (r-1) + (c-1) + (r-1)(c-1)$. Mean squares are calculated as $MS = \frac{SS}{df}$, leading to test statistics $F_{\text{row}} = \frac{MSA}{MSE}$ and $F_{\text{col}} = \frac{MSB}{MSE}$.

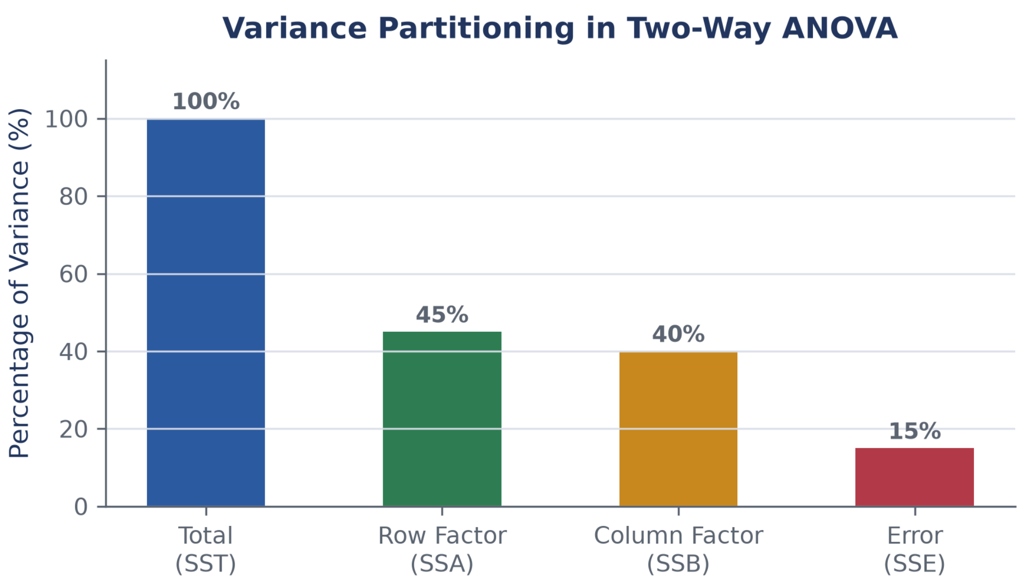

Figure 1: ANOVA divides total dataset variation into row factor, column factor, and residual error components.

Fully Worked Example

Consider a study with 3 rows and 4 columns. Suppose $SSA = 120$, $SSB = 180$, and $SSE = 60$.

Step 1: Calculate degrees of freedom: $df_A = 3 - 1 = 2$, $df_B = 4 - 1 = 3$, $df_E = 2 \times 3 = 6$.

Step 2: Calculate Mean Squares:

$$MSA = \frac{120}{2} = 60$$
$$MSB = \frac{180}{3} = 60$$
$$MSE = \frac{60}{6} = 10$$

Step 3: Compute F-ratios: $F_{\text{row}} = \frac{60}{10} = 6.0$, $F_{\text{col}} = \frac{60}{10} = 6.0$.

2 Covariance and Karl Pearson Correlation

Measure the direction and strength of linear relationship between two variables.

Covariance measures whether two variables move together in the same direction. Because covariance depends on measurement scale, Karl Pearson standardized it into correlation coefficient r, which ranges strictly from -1 to $+1$.

Everyday Picture & Intuition

Covariance is like measuring combined height and weight in inches and pounds; the raw number changes if you switch to centimeters and kilograms. Correlation converts this number to a standard score between -1 and $+1$.

2.1 Formula for Karl Pearson's r

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}} = \frac{\text{Cov}(X, Y)}{s_X s_Y} \tag{2}$$

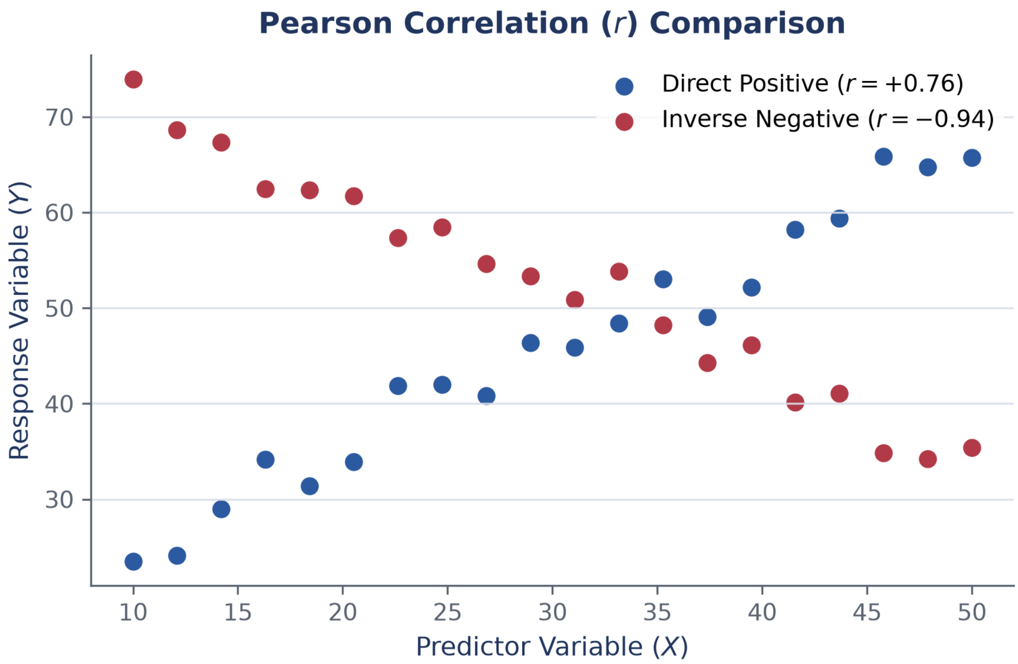

Figure 2: Scatter plot showing strong positive correlation versus strong negative correlation.

Fully Worked Example

Let us evaluate student anxiety (X) versus test scores (Y) across 5 students:

Student	X	Y	$(X - \bar{X})$	$(Y - \bar{Y})$	$(X - \bar{X})(Y - \bar{Y})$
1	2	90	-3	+20	-60
2	4	80	-1	+10	-10
3	5	70	0	0	0
4	6	60	+1	-10	-10
5	8	50	+3	-20	-60

Step 1: Calculate means: $\bar{X} = 5$, $\bar{Y} = 70$.

Step 2: Sum of cross-products: $\sum(X - \bar{X})(Y - \bar{Y}) = -140$.

Step 3: Sum of squared deviations: $\sum(X - \bar{X})^2 = 20$, $\sum(Y - \bar{Y})^2 = 1000$.

Step 4: Compute r:

$$r = \frac{-140}{\sqrt{20 \times 1000}} = \frac{-140}{\sqrt{20000}} = \frac{-140}{141.42} = -0.99 \tag{3}$$

Conclusion: Nearly perfect strong negative correlation. Higher anxiety correlates with lower test scores.

3 Simple Linear Regression Analysis

Predict a continuous target variable using a straight-line trend model.

Simple linear regression models the target variable Y as a linear function of predictor X plus random noise: $Y = a + bX + \epsilon$.

Everyday Picture & Intuition

Think of simple regression as drawing a baseline path through scattered data points so that the total missed vertical distances are as small as possible.

3.1 Least Squares Formulas

Slope coefficient b and intercept a minimize squared errors:

$$b = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2} \tag{4}$$

$$a = \bar{Y} - b\bar{X} \tag{5}$$

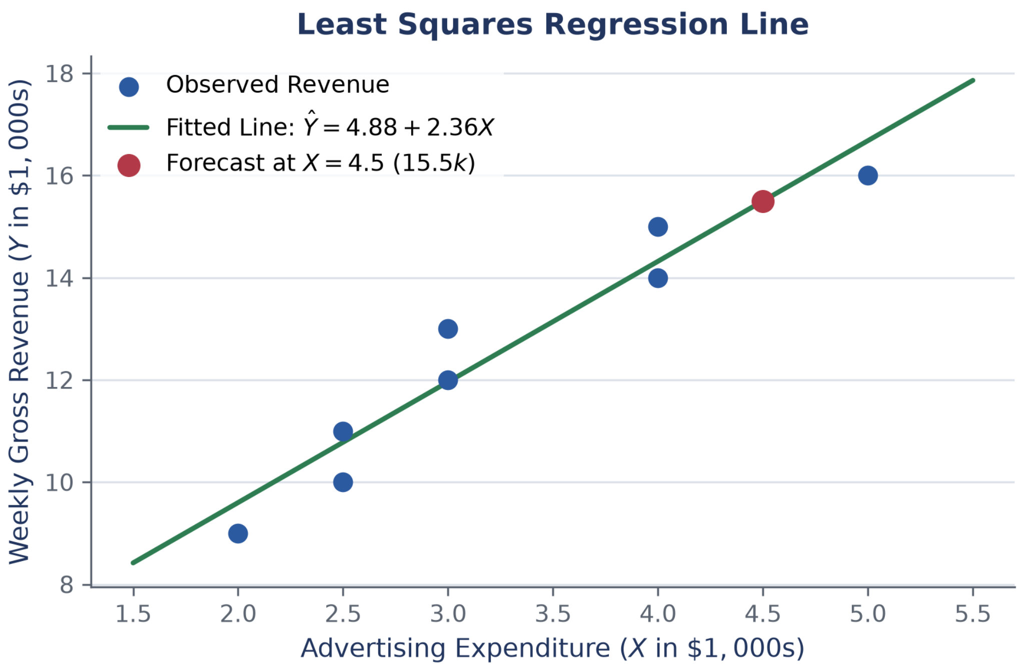

Figure 3: Fitted least-squares regression line estimating weekly movie gross revenue from advertising spend.

Fully Worked Example

A theater owner records advertising expenditure (X in \$1,000s) and weekly gross revenue (Y in \$1,000s): Sample points (X, Y): $(2, 9), (5, 16), (4, 15), (2.5, 11), (3, 13), (4, 14), (2.5, 10), (3, 12)$. Total sample size $n = 8$, $\bar{X} = 3.25$, $\bar{Y} = 12.50$. Given $\sum(X - \bar{X})^2 = 8.50$ and $\sum(X - \bar{X})(Y - \bar{Y}) = 20.00$.

Step 1: Calculate slope b:

$$b = \frac{20.00}{8.50} = 2.353 \approx 2.36 \tag{6}$$

Step 2: Calculate intercept a:

$$a = 12.50 - (2.353 \times 3.25) = 12.50 - 7.647 = 4.853 \approx 4.85 \tag{7}$$

Step 3: Fitted regression model: $\hat{Y} = 4.85 + 2.36X$.

Step 4: Forecast revenue for ad spend $X = 4.5$ (\$4,500):

$$\hat{Y} = 4.85 + 2.36(4.5) = 4.85 + 10.62 = 15.47 \quad \text{(or \$15,470)} \tag{8}$$

Watch Out! Pitfalls

Correlation does not mean causation! High correlation between two variables does not prove that changing one directly causes changes in the other.

Key Takeaways

- Two-Way ANOVA isolates row and column effects simultaneously.
- Karl Pearson's r measures linear association strength on a scale from -1 to $+1$.
- Least-squares regression minimizes squared vertical errors to yield optimal prediction line parameters a and b.